

Enhancing fine-grained sentiment analysis for implicit texts on social media[☆]

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ABSTRACT

Sentiment analysis of social media texts is crucial for understanding the psychological dynamics of internet users and mitigating the potential risks associated with public opinion online. However, the lack of clearly defined sentiment words in many social media texts adversely affects the precision and applicability of sentiment analysis models. To improve the identification of implicit sentiment in social media texts, we propose a RoBERTa-based Fine-Grained Sentiment Analysis model RFGSA. First, the model generates word vectors using the RoBERTa model. Subsequently, context and local features are extracted using BiLSTM and TextCNN, respectively. To address the limitations of traditional multi-feature fusion methods, we propose an adaptive feature fusion approach based on GRU. This method dynamically adjusts feature fusion parameters according to the underlying data characteristics and model requirements, thereby improving the model's accuracy. The experimental results show that on the WASSA 2018 IEST implicit sentiment dataset, the proposed model improves the accuracy and F1 score by 1.45% and 1.4%, respectively, compared to the best baseline model, Amobee. Additionally, the proposed model improves the accuracy and F1 score by 4% and 4.59%, respectively, over the DuFER model. In addition, on the Semeval 2018 explicit sentiment dataset, the accuracy and F1 score of the proposed model are similar to those of the DuFER model, with the difference being less than 0.2%. This result verifies the proposed model's good generalization ability and its wide application potential in multi-task environments.

1. Introduction

In the era of Big Data, internet users frequently share their views on various social media platforms, leading to a significant increase in the volume of texts. This surge in data provides a valuable resource for sentiment analysis research. The purpose of sentiment analysis is to use natural language processing technology to identify the sentiment tendencies in the texts (Wankhade et al., 2022). The application of this technology has a wide range of cross-field influence. In the commercial field, the technology can extract the text and social behavior characteristics of user reviews, accurately mine consumers' implicit attitudes toward products, and guide the optimization of marketing strategies. In the field of public safety, the technology is used to analyze social media comment data and identify group mood swings to support emergency response. In politics, the technology can identify the public's emotional predisposition to news texts to assess the public's acceptance of policies. Therefore, sentiment analysis is an academic research

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problem and an important tool for decision support in many fields. With the continuous progress of natural language processing technology, sentiment analysis will play an important role in more fields and promote information exchange and understanding at all levels of society.

The sentiment analysis research was first started in 2002. In 2013, researchers proposed the word embedding model, which also promoted the development of sentiment analysis research to a certain extent. After 2018, BERT, RoBERTa, and other pre-trained language models were proposed successively, and the application of these models can significantly improve the performance of sentiment analysis tasks. Sentiment analysis is typically classified into two categories based on the presence or absence of explicit sentiment features in the texts: Explicit Sentiment Analysis (ESA) and Implicit Sentiment Analysis (ISA). ESA refers to extracting words or phrases directly expressing sentiment polarity from the text (Budhi et al., 2021). ESA predominantly uses explicit sentiment words to determine sentiment polarity, such as happy, sad, angry, satisfied, etc. The advantage of this method is that it is simple in structure and can extract emotional information from text quickly and efficiently. However, its accuracy is easily affected by the polysemy of emotional words, contextual changes, and complex expressions. ISA is more focused on the indirect expression of emotion inferred from the text, such as the author's intention, metaphor, irony, background knowledge, and other aspects to convey emotion indirectly. For example, although the weather was bad, we still went to climb the mountain. The sentence does not show obvious emotional vocabulary, but from the context, it can be inferred that a positive emotional tendency. ISA presents greater challenges compared to ESA.

With the gradual increase in implied sentiment texts by internet users of social media platforms (Ocampo et al., 2023), ISA has become an important topic in natural language processing. Existing sentiment analysis generally uses rule-based methods. With the development of deep learning, researchers have begun applying deep learning models to analyze sentiment characteristics embedded within implicit expressions. For example, some researchers use BiLSTM to capture global semantic information and aspect category information (Wu et al., 2021; Yang et al., 2021; Shuang et al., 2020; Tan et al., 2022). Additionally, some researchers design ISA models based on the CNN model to extract local features of implicit sentiment expressions (Wu et al., 2021; Wei et al., 2020). In recent years, Transformer models have significantly enhanced the performance of sentiment analysis. In addition, feature fusion technology plays an important role in sentiment analysis tasks. It can significantly improve the performance of sentiment analysis models through the fusion of multiple feature representations. Ahmed proposed a feature fusion framework based on convolutional neural networks (CNN), which effectively improved the performance of sentiment analysis of short texts through local feature extraction and multi-level feature fusion (Abas et al., 2021). Liang proposed a feature fusion method based on graph convolutional networks (GCN), which enhanced the model's ability to understand complex emotional semantics by modeling the relationship between features (Liang et al., 2022).

Although existing research has achieved some progress in sentiment analysis, several issues remain to be addressed. On the one hand, implicit sentiment texts generally lack explicit sentiment words and use complex linguistic features such as metaphor and irony, which do not directly convey sentiments. Traditional deep-learning methods often perform poorly in capturing and understanding these subtle sentiment cues. On the other hand, ISA often requires the integration of multiple characteristics. However, most fusion methods remain static. They typically combine different feature types through simple splicing. The challenge with these methods is that they do not account for the varying contributions of each feature. Static fusion assumes equal importance for all features, overlooking their interactions. This limitation may hinder the model's performance. To address these issues, we propose a RoBERTa-based sentiment analysis model, RFGSA. This model analyzes both explicit and implicit texts on social media for sentiment characteristics and implements a fine-grained classification of the sentiment of the texts. Firstly, we preprocess text from social media posts, including adding data labels, stopping words, word segmentation, etc. Secondly, word vector features are extracted based on RoBERTa. These word vectors are then fed into the neural network models to extract the context and local features of the texts. Next, an adaptive feature fusion module is designed for feature fusion. Finally, the fused features are fed into a Softmax classifier for fine-grained sentiment classification.

The main contributions of this paper are as follows:

- Inspired by existing research on ESA and ISA, we propose a RoBERTa-based Fine-Grained Sentiment Analysis model, RFGSA. This model analyzes both explicit and implicit texts on social media for sentiment characteristics, implements a fine-grained classification of the sentiment of the texts, and improves the model's generalization and adaptability;
- We propose an adaptive feature fusion approach based on the Gating Recurrent Units (GRU-FUSION). This approach can dynamically adjust the feature fusion parameters according to the feature of the underlying data and the requirements of the model, thereby improving the accuracy of the model;
- After rigorous experimental verification, the proposed model shows significant advantages over existing ISA models in several key performance indicators, including accuracy, recall rate, and F1 score.

The article is structured as follows: Section 2 reviews the related work, Section 3 introduces the proposed model, Section 4 presents the experimental setup and results, and Section 5 concludes the article.

2. Related work

Feature extraction is the basis of sentiment analysis, and its precision will directly affect the results of the subsequent effective analysis. Given the differences in the principle of feature extraction, this paper divides the existing research into the following three categories:

2.1. Feature extraction method based on rule

Rule-based feature extraction methods use predefined, explicit logical rules or patterns to identify and construct new features directly from the original data (Hearst, 1992; Sarawagi, 2008). Their advantages are high interpretability and direct use of domain knowledge. They are suitable for scenarios with clear patterns, rules, or high transparency of feature sources.

Rule-based sentiment analysis mainly uses prior knowledge and linguistic rules (Vasileios et al., 1997; Bo et al., 2002; Mingqing et al., 2004), including an emotional dictionary, grammar, syntactic structure, etc. The advantage of this method is that it is interpretable and does not require a large amount of annotation data. Zhang proposes an innovative approach that combines rule-based strategies with deep learning models to improve the accuracy and robustness of sentiment analysis models (Zhang et al., 2015). Based on the framework of this research, the method has been widely used and has made remarkable progress in recent years. Liao et al. proposed a two-stage fine-grained text sentiment classification model, which used syntactic rule matching and depth semantics (Liao et al., 2021). The model demonstrated good performance on public datasets. Paramonov et al. proposed a sentiment analysis method based on Russian semantic rules, which replaced the original list of words with a sentence syntax tree (Paramonov and Anatoliy Poletaev, 2021). Anuratha et al. proposed a SLFEM-L-2-S2RRNN based on the model of social spider lexical feature set (SLFEM-L-2-S2RRNN) (Anuratha and Parvathy, 2022). Rahab used an optimized heuristic, an Arabic opinion corpus, and a third rule model to classify sentiment in Arabic (Rahab et al., 2023). Saraswathi et al. proposed a feature selection method for language rules to classify the unlabeled dataset obtained online (Saraswathi et al., 2023). Diwali et al. proposed a rule-based method for sentiment analysis, which combined the rules of dependency syntax with a deep neural network model to realize the sentiment classification (Diwali et al., 2022). Li proposed a rule-based transformation strategy to enhance the information in the tag space, used the word-level sentiment distribution vector (WED) to generate new tags, and added them to the original tags, thus the category of the sentiment classification task was extended (Zongxi et al., 2021). Michael combined machine learning with a rule-based method for sentiment analysis of Chinese blog content (Chau et al., 2020).

However, rule-based sentiment analysis methods mainly use predefined sentiment dictionaries and linguistic rules, such as positive and negative classification of sentiment words. It is generally able to recognize explicitly expressed sentiment but does not perform well on implicit sentiment expressions. Since rules are mainly based on implicit expressions of sentiment, it is impossible to fully understand the complex semantic and contextual relationships in the text. Therefore, for texts containing implicit sentiment, rule-based methods often fail to accurately capture the true meaning of sentiment, and their performance is relatively limited.

2.2. Feature extraction method based on machine learning

Feature extraction methods based on machine learning use algorithms to learn from data and generate informative feature representations automatically (Hema et al., 2023; Yerramaneni and Reddy, 2025). It is better at dealing with High-dimensional, complexity and unstructured data. Compared with rule-based methods, machine learning methods can automatically learn complex patterns and have better generalization potential.

The method based on Machine Learning utilizes a large amount of labeled text data for training, allowing it to learn sentiment characteristics from the data. Both the Machine Learning Model and Deep Learning Model demonstrate strong performance in analyzing large text datasets, and sentiment analysis usually uses Machine Learning or Deep Learning to extract sentiment features from the text. For example, some models utilize BiLSTM and TextCNN to extract sentiment features from textual data. These models can capture more comprehensive sentiment information from the text thereby improving the model's accuracy.

Ananthajothi used a neural network (NN) algorithm to train implicit sentiment data while using an adaptive beetle swarm optimization algorithm (SA-BSO) for feature selection (Ananthajothi et al., 2022). Basiri proposed an Attention-based Bidirectional CNN-RNN Deep Model(ABCDM), which is based on BiLSTM, GRU, and Attention Mechanism to extract text context information, and used convolution and pooling operations to extract local text part features (Basiri et al., 2021). Wu proposed an approach based on the Attention Mechanism and Bidirectional Long-Short-Term Memory Network (SC-ABiLSTM) for processing the sentiment classification task of implicit short-text expression (Wu et al., 2021). Yang proposed an ISA model based on a Graph Attention Convolutional Neural Network (GACNN). The model used a graph convolutional neural network to spread semantic information and adopted the attention mechanism to calculate the importance of each word to the sentence (Yang et al., 2021). Gao proposed a method based on CNN and BiGRU for short-text aspect sentiment analysis. The hybrid model could fully extract text features and address the issue of long-distance dependence in sequences (Gao et al., 2022). Wei proposed an ISA model based on BiLSTM and the Multipolar Attention Mechanism. The proposed Multipolar Attention Mechanism could recognize the difference between word and sentiment orientation, at the same time, an orthogonal constraint mechanism was adopted to ensure that classification performance is maintained during optimization (Wei et al., 2020). Shuang proposed a Feature Distillation Network (FDN), designed a two-door mechanism to realize the interaction between the aspect and its corresponding context, and solved the noise problem in affective analysis, improved the accuracy of affective profiling (Shuang et al., 2020). Gan proposed a model of a multi-channel convolutional neural network and attention mechanism, which used a multi-channel structure to capture different high-level semantic features, and through local attention and global attention to enhance the model's ability to classify sentiment (Gan et al., 2021). Li used Internet slang and emoticons as features of sentiment analysis and proposed an Attention Mechanism-based Bidirectional Long-Short-Term Memory Recurrent Neural Network model (ATTBiLSTM) to implement sentiment classification of microblogs (Li et al., 2020). Fu used BiLSTM to capture global semantic information and aspect category information and focused on aspect category and sentiment information through multiple attention mechanisms, this reduced the problem of interference between different aspects of sentiment information (Fu et al., 2021).

Traditional machine learning methods such as BiLSTM and TextCNN perform well in sentiment analysis tasks, efficiently extracting sentiment features from large amounts of labeled text and extracting explicit sentiment information from local context. These methods are effective when dealing with intuitive, explicit expressions of emotion. However, these methods face challenges in addressing word ambiguity and performing fine-grained sentiment analysis. They often struggle with polysemy, where a single word can carry multiple meanings depending on the context, and may fail to capture the subtle sentiment nuances embedded in different parts of a sentence. Therefore, solely relying on these methods results in poor performance in ISA tasks.

2.3. Feature extraction method based on transformer

The Transformer model architecture lays the foundation for large-scale pre-training language models such as BERT, RoBERTa, GPT, etc. These Transformer-based models significantly improve the performance of natural language processing tasks by pre-training on large-scale text data and then fine-tuning it for specific tasks.

Jacob Devlin first proposed the BERT (Bidirectional Encoder Representations from Transformers) model based on Transformer (Devlin et al., 2019), this model can greatly improve the performance of sentiment analysis. After that, researchers conducted a lot of studies on the application of BERT in sentiment analysis (Mickel et al., 2019; Songwen et al., 2019; Abdul et al., 2019). BERT-based sentiment analysis still has problems of insufficient optimization of training task Settings and low computational efficiency. To solve these problems, researchers proposed the RoBERTa model, which improved the BERT model by removing the NSP task, adding training data, optimizing mask strategy, and other methods. This makes the model perform better in tasks such as sentiment analysis and improves computational efficiency and robustness (Liu et al., 2019; Wenxiong et al., 2021). Sentiment analysis models based on BERT and RoBERTa have excellent performance in understanding context and capturing the relationship between words. However, they mainly use Transformer architecture to process the context information of long texts, which makes it easy to lose the capture of local emotion information in long texts. In addition, it performs poorly on tasks such as fine-grained sentiment analysis and capturing local emotion patterns in long texts. Therefore, researchers combine Transformer with LSTM and CNN models for emotion analysis. Some researchers use pre-trained Transformer models BERT or RoBERTa to extract text features and combine them with other neural network structures (e.g. RNN, GCN) to handle complex sentiment interactions or enhance the performance of sentiment analysis (Cheruku et al., 2024; Tan et al., 2022; Xiong et al., 2023; Lu et al., 2022). Some researchers also use multi-task learning frameworks or feature fusion mechanisms to deal with both ESA and ISA tasks (Cai et al., 2023; del Arco et al., 2022; Zhang et al., 2021). They are suitable for dealing with complex sentiment tasks, especially when explicit and implicit sentiment are present at the same time, and these methods can enhance the comprehensive performance of the model. Khoshnam proposes a dual framework for identifying implicit and explicit emotions. Based on computational linguistics methods and BERT models, subtle sentiment tendencies are captured to gain insight into the contextual semantics of text (Khoshnam and Baraani-Dastjerdi, 2022).

Inspired by the above research, we propose a RoBERTa-based sentiment analysis model RFGSA, which fully considers the poor adaptability of multi-feature fusion methods and proposes an adaptive feature fusion method based on GRU. The model combines RoBERTa, deep learning techniques, and attention mechanisms to analyze fine-grained sentiment features of explicit and implicit text on social media.

3. Proposed methodology

3.1. Overview of the model

As shown in Fig. 1, the RFGSA model consists of five key components:

- (1) Input Layer: First, the model preprocesses social media texts, including adding data labels, removing stop words, and segmenting words, among other tasks. The processed texts are fed into the RoBERTa model to generate word embeddings.
- (2) Local Feature Extraction Layer: The embeddings generated by RoBERTa are fed into a TextCNN model. TextCNN is designed to extract local features from these embeddings by utilizing convolutional filters of various sizes to capture n-gram features.
- (3) Context Feature Extraction Layer: The embeddings generated by RoBERTa are fed into a BiLSTM model. The BiLSTM captures long-range dependencies and contextual information by processing the texts in forward and backward directions.
- (4) Feature Fusion Layer: The feature weights are dynamically adjusted based on the GRU to reflect the contribution of different features. Local features extracted by TextCNN and contextual features obtained from BiLSTM are then integrated to form a comprehensive text representation, leveraging local and global information.
- (5) Output Layer: The fused features are fed into the full connectivity function and Softmax function to predict the sentiment category of social media texts.

3.2. Input layer

The input layer focuses on the vectorized representation of the texts. The texts will be preprocessed before the vectorized representation of the collected social media texts, including noise removal (e.g., URLs and special characters), word splitting, stop words removal, and de-duplication. Then, we use the RoBERTa model to extract word sequences and convert them into contextual word vectors.

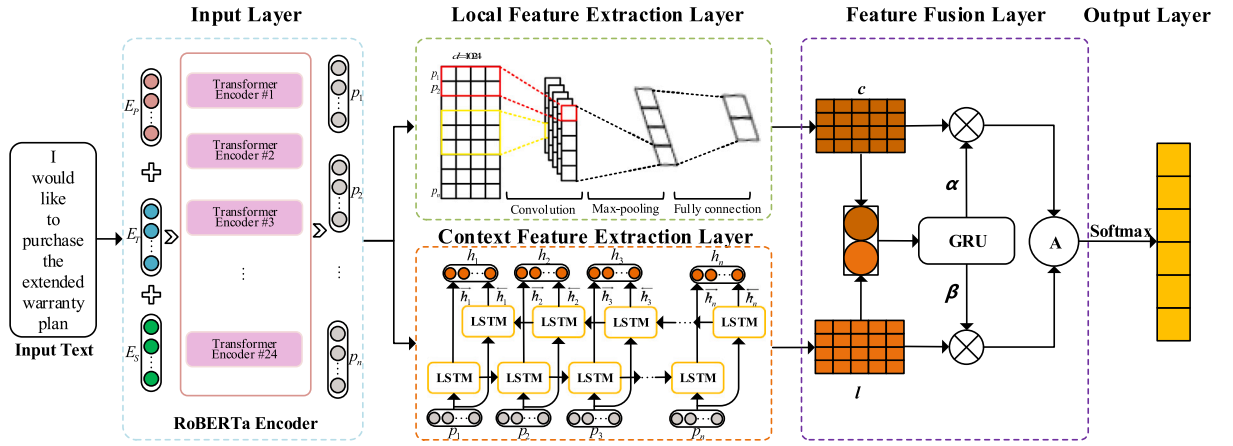


Fig. 1. Framework of RFGSA. The number of tokens in the input text is represented by n . In the input layer, E_p represents positional embeddings, E_w represents word embeddings, E_s represents segment embeddings, and p represents output vectors. Each output vector has dimension $d_p = 1024$. In the context feature extraction layer, h represents the hidden layer feature of BiLSTM and feature dimension $d_h = 1024$. In the local feature extraction layer, the convolution layer adopts one-dimensional convolution and max pooling layer, and the filter size table is (2, 3, 4). The full connection layer output vector dimension is 1024.

RoBERTa is a pre-trained language model based on the Transformer framework, which is optimized based on the BERT model by removing the Next Sentence Prediction (NSP) task in BERT. It also increases the diversity of the trained models by adopting the dynamic masking technique. This technique randomly performs the masking operation on the input when entering the model. These improvements enable RoBERTa to enhance its ability to capture complex interactions and dependencies between words, recognize subtle mood changes, and provide high-quality word embedding for downstream tasks (Liu et al., 2019).

The input of the RoBERTa consists of Token Embeddings, Position Embeddings, and Segment Embeddings. Token Embeddings capture the semantics of the words or subwords in the input text by mapping them to a high-dimensional vector space. Position Embeddings is a fixed embedding that represents the index of each position in the sequence and enables the model to understand and distinguish words at different positions. Segment Embeddings are used when dealing with tasks containing multiple paragraphs or text fragments, and paragraph identifiers differentiate between different paragraphs or fragments. In RoBERTa, the [CLS] token is added at the beginning of an input sentence, and its output vector is used to capture the global representation of the entire sentence. It is typically employed for text classification tasks. The [SEP] token is commonly used as a separator marker. For single-sentence inputs, it is positioned at the end of the sentence. Its primary function is to help the model delineate the boundary of the sentence. The input of the RoBERTa is shown in Fig. 2. In the pre-training phase, the RoBERTa's main task is masked language modeling (MLM). RoBERTa randomly selects 15% of the words in the input text as masks. Of these, 80% are substituted with [MASK] tokens, 10% are substituted with randomly chosen words, and the remaining 10% remain unchanged. This strategy enables the model to learn to process different contextual information and improves the model's ability to predict real words. However, its masking strategy is different from the BERT model. RoBERTa introduces a dynamic masking mechanism that dynamically adjusts the mask position in each training batch, which means that the same vocabulary list can be masked in different ways in different training iterations. This method enhances the robustness and generalization of the RoBERTa. In addition, RoBERTa uses larger batches and learning rates during training, which allows it to converge faster and better utilize computational resources. By performing this task, RoBERTa generates pre-training representations with strong semantic comprehension, providing accurate and rich contextual information for downstream tasks. As shown in Fig. 3, the RoBERTa consists of an Input Layer, Transformer Layer, and Output Layer. In the Input Layer, RoBERTa generates Token Embeddings, Position Embeddings, and Segment Embeddings according to the described method. It then combines these embeddings to create the initial input vectors for each token. The input vectors are then fed into the Transformer layer. In our proposed model, the RoBERTa is set up with a 24-layer Transformer bi-directional encoder to extract word embedding features. In each layer, the Transformer processes the input vectors through the self-attention mechanism and feed-forward neural networks, which enable the model to pay attention to the input at different positions and learn the complex dependencies between words, thus providing richer feature representation. In the Output Layer, the vectors processed by the Transformer are pooled, and the last layer's hidden state is selected as the final representation of the RoBERTa. The output vectors are represented as $p = (p_1, p_2, \dots, p_i, \dots, p_n)$, where $p_i \in \mathbb{R}^d$ denotes the word embedding feature of the i th word, d denotes the dimension of the vectors, and n denotes the length of the vectors.

3.3. Local feature extraction layer

In social media texts, implicit expressions of sentiments are often conveyed through subtle changes in phrasing and local features. The pre-trained word embedding in Transformer (e.g., BERT, RoBERTa) can integrate global context information to solve the polysemy ambiguity problem of traditional static word vectors (e.g., Word2Vec). For example, the different meanings of “apple” in “iPhone” and “eat apple” can be distinguished by dynamic embedding. In addition, it can also provide more fine-grained semantic

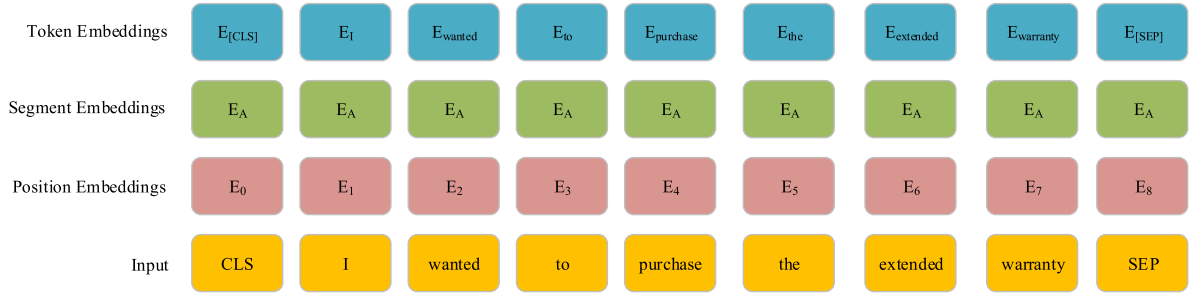


Fig. 2. Input of RoBERTa model.

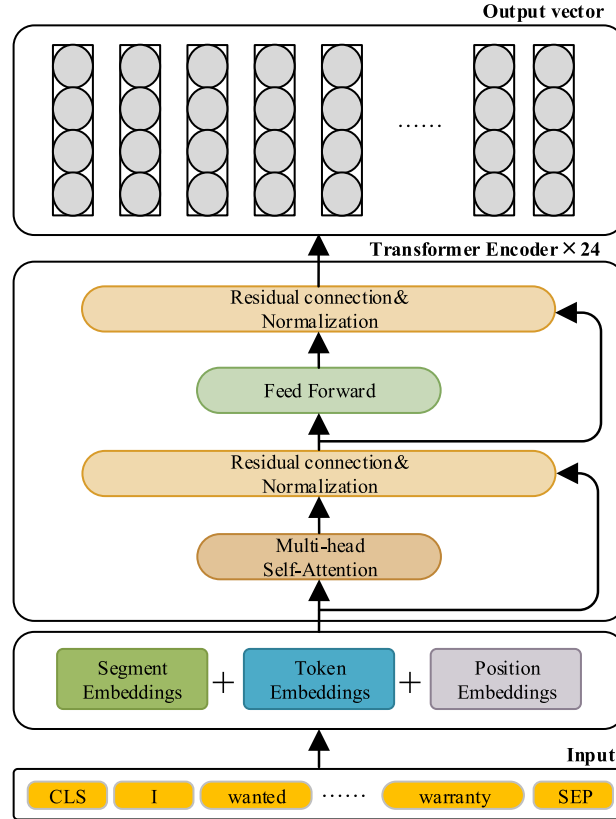


Fig. 3. Framework of RoBERTa model.

features. However, if the Transformer is directly used for local feature extraction, multiple layers need to be stacked, and the calculation cost is high. To address this issue, we propose a strategy that combines RoBERTa with a local feature extraction module. By integrating RoBERTa with this module, we can leverage the contextual features offered by RoBERTa while compensating for its deficiency in local feature extraction. This combined approach aims to enhance the accuracy and comprehensiveness of implicit emotion analysis.

TextCNN is a convolutional neural network model for processing textual data. It maps words in texts into a high-dimensional space. Then, it performs convolutional operations on these word vectors using multiple convolutional kernels of different sizes. This process captures the text efficiently by capturing local (i.e., phrase-level) semantic features. TextCNN is relatively simple with low computational complexity (Luo, 2022). Our proposed model takes the embedding vectors output from RoBERTa as input to the TextCNN for extracting local features of the texts. The specific structure is as follows:

(1) Convolution layer

The convolution layer extracts local features through convolution operations. In general, text convolutional neural networks (TextCNN) ensure the comprehensive capture of text features through multiple convolutional layers. Each convolution layer can use filters of different sizes (convolution kernel) to process the same text, thus learning more and different feature information. Suppose

a $d \times k_j$ filter is used to extract the local features of the sentence p , and the result obtained after the convolution operation is:

$$c_{ij} = \text{Relu}(w_j * p_{i:i+k_j-1} + b_j) \quad (1)$$

where the convolution kernel is $w_j \in \mathbb{R}^{d \times k_j}$, $j = 1, 2, \dots, m$ denotes the j th convolutional kernel. d denotes the word vector dimension, k_j denotes the size of the convolution kernel. $p_{i:i+k_j-1}$ denotes the feature matrix.

In this study, a STEP length of 1 is applied, and three convolution kernels with different sizes (2, 3, 4) are utilized to capture the local features corresponding to 2-g, 3-g, and 4-g segments of the sentences, respectively. To accelerate the convergence process of training, we choose *Relu* function as the activation function, and its mathematical expression is shown as follows:

$$f(x) = \max(0, x) \quad (2)$$

where, when $x \geq 0$, $f(x) = x$, which is the input value itself. When $x < 0$, $f(x) = 0$. In the neural network model, an activation function is introduced to capture the nonlinear relationship inside the network. The commonly used activation functions are *Sigmoid*, *tanh*, *ReLU*, and so on. The introduction of an activation function helps the network to learn and express complex nonlinear features. The main reason for choosing *ReLU* as the activation function is that it has a derivative of 1 in the positive region is that it can effectively solve the gradient disappearance problem and provide a fast convergence rate during training. Compared to functions such as *Sigmoid* and *tanh*, *ReLU* is easy to compute and outputs positive inputs directly, helping to speed up the learning process. *ReLU* has become a common activation function in deep learning.

After the filter convolves the sentence p from top to bottom with step = 1, we can get a set of local vectors, which can be expressed as:

$$c = \{c_1, c_2, \dots, c_{i+k_j-1}\} \quad (3)$$

(2) Pooling layer

The main purpose of the pooling layer is to downsample features in the convolutional neural network, which helps to reduce the number of parameters, reduce the dimension of features, prevent overfitting, and enhance the robustness of the model. At present, the two common pooling methods are maximum pooling and average pooling. Average pooling is the calculation of the average for each pooling area. Maximum pooling is the selection of the maximum value from each pooling region as the output for that region, so it is better to capture the most prominent features within the region.

We apply the maximum pooling method to pool the local feature maps obtained through the convolution operation, extracting the maximum feature value from each pooling region, which can be expressed as follows:

$$\hat{c} = \text{Max}(c) \quad (4)$$

(3) Fully Connected Layer

The main purpose of the Fully Connected Layer (FC layer) is to further integrate the features extracted from the convolution layer and the pooling layer. At the fully connected layer, we merge all $\hat{c}$ as the final output vector of the TextCNN.

3.4. Context feature extraction layer

In natural language, semantics is characterized by obvious continuity, and the order of occurrence of words can often influence the meaning of a sentence (Mutlu and Sezer, 2023). To enhance the model's sequential modeling capability and capture subtle changes and contextual information in social media implicit texts, we introduce the BiLSTM model to extract the contextual features of the texts. The BiLSTM consists of two LSTM layers in opposite directions so the model can more accurately capture the contextual information of the sequence. It effectively handles contextual features, especially in the case of the lack of obvious words in the implicit expression, demonstrating strong performance (Wei et al., 2020). Transformer can rely on self-attention to capture remote dependencies and generate very rich contextual information. However, for ISA tasks or other tasks that require stepwise context modeling, word embeddings relying solely on the Transformer's pre-trained model may not be sufficient to accurately capture sequence timing and local dependencies. Therefore, BiLSTM's step-by-step processing capabilities and sequential modeling features are critical for certain fine-grained tasks like ISA.

We use the vectors obtained from the training of the RoBERTa as inputs to the BiLSTM model, where the forward LSTM obtains the result $\vec{h}_i$ through the oblivious gate layer, the input gate layer, the memorized state layer, and the output gate layer, and $\overleftarrow{h}_i$ in the reverse LSTM layer. The main formulas are as follows:

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci}c_{t-1} + b_i) \quad (5)$$

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf}c_{t-1} + b_f) \quad (6)$$

$$\tilde{c}_t = \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_{bc}) \quad (7)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (8)$$

$$o_t = \sigma(W_{xo}x_t + W_{co}h_{t-1} + W_{co}h_t + b_o) \quad (9)$$

$$h_t = o_t \odot \tanh(c_t) \quad (10)$$

where x_t is the input of time step t , h_{t-1} is the hidden state of the last time step, and c_{t-1} is the cell state of the last time step. i_t , f_t , $\tilde{c}_t$, c_t , o_t are input gate, forgetting gate, cell state, output gate, and hidden state, respectively. W and b are model parameters, σ is sigmoid function, $\tanh$ is hyperbolic tangent function. For the input $\tilde{c}_t$ at time t , the obtained hidden state representations $\tilde{h}_t$ and $\overleftarrow{h}_t$ are shown in the following equations, respectively:

$$\tilde{h}_t = \overrightarrow{LSTM}(p_t), t \in [1, n] \quad (11)$$

$$\overleftarrow{h}_t = \overleftarrow{LSTM}(p_t), t \in [1, n] \quad (12)$$

The output of a combination of forward and reverse outputs is expressed as follows:

$$h = \tilde{h}_t \oplus \overleftarrow{h}_t \quad (13)$$

where h is the final output vectors of the context feature extraction layer, represented by l in this paper.

3.5. Feature fusion layer

Existing research on sentiment analysis usually has simpler methods for multi-feature fusion, such as feature splicing, weighted summation, and element-level product. However, such methods do not consider the consistency of features (Xiong et al., 2024). Specifically, for different types of textual tasks, the degree of contribution of local and contextual features to the results may be inconsistent. Simple fusion methods can lead to conflicting information, affecting the fused features' quality and limiting the model's performance.

To address this issue, we propose an adaptive feature fusion approach based on the GRU, which adjusts the degree of feature fusion and enhances the final fusion effect by learning the weight parameters. GRU is a type of Recurrent Neural Network (RNN) architecture introduced to address the limitations of traditional RNNs, particularly in capturing long-term dependencies and mitigating the vanishing gradient problem. GRU itself is widely used in many deep learning tasks, especially in time series data processing and feature fusion, but the dynamic regulation mechanism of GRU is not fully reflected in many sentiment analysis tasks, especially ISA tasks. Therefore, our approach introduces a distinctive perspective in ISA tasks. In particular, when addressing contextual and local features via GRU, it allows for an adaptive adjustment of fusion weights, ensuring a balanced representation of both emotions.

The overall architecture of the Feature Fusion Layer involves first concatenating the features c and l . These concatenated features are then separately input into GRU to compute the update gate, reset gate, and candidate state, generating the feature weights α and β for c and l , respectively. Next, the features c and l are fused using these weights. Finally, the fused features are passed through a self-attention mechanism, with the output of this mechanism serving as the final output of the Feature Fusion Layer. The detailed process of feature fusion is described below.

First, we concatenate the local feature c and the contextual feature l generated in the previous step to obtain the feature $[c : l]$, which is then fed into the GRU for weight calculation. The weight of the local feature c is calculated as follows:

$$z_c = \sigma(W_z \cdot [c : l] + b_z) \quad (14)$$

$$r_c = \sigma(W_r \cdot [c : l] + b_r) \quad (15)$$

$$\tilde{h}_c = \tanh(W_h \cdot [r_c \odot c, l] + b_h) \quad (16)$$

$$\alpha = (1 - z_c) \odot c + z_c \odot \tilde{h}_c \quad (17)$$

where r_c is the output of the reset gate, W_r is the weight matrix of the reset gate, $[c : l]$ is the input at the current moment, and b_r is the bias term of the reset gate. z_c is the output of the update gate, and W_z is the weight matrix of the update gate. b_z is the bias term of the update gate. h_t is the candidate's hidden state. W_h is the weight matrix of the feature h , and b_h is the bias term. α is the vector of weights of the final output of c at the current moment.

Similarly, the weight matrix β of the contextual feature l is computed by this method. Then, we weight the context feature l and the local feature c with the attention weights obtained from the previous training. The formulas are as follows:

$$\hat{c} = \alpha \cdot c \quad (18)$$

$$\hat{l} = \beta \cdot l \quad (19)$$

Next, we designed the bilinear function to extract the interrelationships between all features:

$$I = \hat{h}^T W_f \hat{c} \quad (20)$$

where W_f is the parameter of the bilinear function, then we design a fully connected network and activation function to realize the fusion of contextual features and local features, $f = \text{ReLU}(W_f[\hat{l} : \hat{c} : I] + b_f)$, where $[\hat{l} : \hat{c} : I]$ denotes the splicing of $\hat{l}, \hat{c}, I$. W_f and b_f are the parameters of the fully connected network, and f is the fused feature.

Finally, we use the Self-Attentional Mechanism to process the fusion features further, enabling the model to focus on important features in the texts and enhance its ability to capture subtle emotional information. The Self-attention Mechanism is calculated as follows:

$$\text{Attention}_{ij} = \text{Softmax}\left(\frac{Q_i \cdot K_j^T}{\sqrt{d_k}}\right) \quad (21)$$

Table 1
The number of tweets.

Dataset	Train	Trial	Test	Total
EID	3,613	342	3,142	7,097
IEST	153,383	9591	28,757	191,731

Table 2
Main parameter description of the model.

Parameter name	Parameter description
Word vector dimension	Represents the feature space size for word embeddings in the model.
Filter sizes	Determines the range of feature extraction in convolution operations.
Filter number	The number of convolution layer filters determines the output depth.
Activation function	Adds nonlinearity to the model for better learning capacity.
Pooling method	Pooling method to reduce the size of feature map.
Hidden layer size	Number of neurons in the hidden layer, controlling complexity.
Learning rate	Controls the speed at which the model weights are updated during training.
Batch-size	Specifies the number of samples per training iteration for optimization.
Dropout	Randomly removes neurons during training to prevent overfitting.
Loss function	Measures the difference between predicted and actual values, guiding optimization.
Optimizer	Updates model parameters based on gradients to minimize the loss.
Output vector length	Defines the size of the prediction vector for final output.

$$\hat{f}_i = \sum_j Attention_{ij} \cdot V_j \quad (22)$$

where Q , K , and V denote the query vector, key vector, and value vector, respectively, which are generated from the input vectors f through linear transformations. $\sqrt{d_k}$ denotes the scaling factor. $\hat{f}$ denotes the output vector.

3.6. Output layer

The above feature vectors are fed into the full connectivity function and Softmax function to predict the sentiment category of social media posting content. The fully connected formula is as follows:

$$z = W_c \hat{f} + b_c \quad (23)$$

where W and b are learning parameters. Finally, z is input into the Softmax function to be converted into a probability distribution:

$$Softmax(z)_i = \frac{e^{z_i}}{\sum_{j=1}^k e^{z_j}} \quad (24)$$

4. Experimental

4.1. Experimental setup

We designed a series of comparative experiments to evaluate the performance of the proposed model using various benchmark data. To train the proposed RFGSA model for fine-grained sentiment classification of explicit and implicit expressions, we evaluated it on two public sentiment analysis datasets: the EID from Semeval 2018 (Mohammad and Bravo-Marquez, 2017) and the WASSA 2018 IEST (Klinger et al., 2018), as shown in Table 1. The EID from the Semeval 2018 dataset is derived from Tweets, which categorize tweets into four types of emotions (Joy, Sadness, Fear, and Anger), and each tweet contains a sentiment label and a label indicating the intensity of the sentiment. The IEST dataset was developed for the WASSA 2018 IEST task, which required the model to predict the sentiment of tweets without explicitly expressing the sentiment word.

The two datasets were randomly divided into a train set, trial set, and test set, and the segmentation ratio was 8:1:1. All experiments were performed on a Linux system with 3090 GPUs assembled. The setting of experimental parameters includes the experimental parameters of RoBERTa, TextCNN, BiLSTM, and FUSION-GRU models. The model parameter descriptions are shown in Table 2, and the experimental parameter values are shown in Table 3.

4.2. Experimental evaluation index

We tested the proposed model on the test set. *Accuracy(A)*, *F1*, *Precision(P)* and *Recall(R)* were used to evaluate the test effect. Accuracy is the percentage of text that is accurately classified. *F1* score is a balanced F-score, defined as the harmonic average

Table 3
Main parameter settings of the model.

Parameter name	Parameter value
Word vector dimension	1024
Filter sizes	2,3,4
Filter number	128
Activation function	ReLU
Pooling method	Max-Pooling
Hidden layer size	1024
Learning rate	1e-5
Batch-size	64
Dropout	0.1
Loss function	Cross entropy
Optimizer	AdamW
Output vector length	2048

between precision and recall, that provides a balanced assessment of model performance. The evaluation indicators are calculated as follows:

$$Accuracy = \frac{TP + TN}{TP + FN + FP + TN} \quad (25)$$

$$Recall = \frac{TP}{TP + FN} \quad (26)$$

$$F1 = \frac{2 \times P \times R}{P + R} \quad (27)$$

where TP and TN represent the number of positive and negative categories correctly predicted, respectively. FP and FN denote the number of positive and negative categories that were incorrectly predicted, respectively. In the experiment, positive categories were used as positive samples and negative categories were used as negative samples to evaluate the model's accuracy in fine-grained emotion classification.

4.3. Experimental results

The evaluation indexes used in the experiment were *Accuracy*(A), $F1$, *Precision*(P) and *Recall*(R). In addition, increasing the number of parameters may potentially improve the model's performance. Therefore, we compared the number of parameters for each model, denoted as 'Parameters.' On the IEST dataset, we compare the performance of our proposed model with the 12 baseline approaches, as shown in Table 4. On the EID from the Semeval 2018 dataset, the performance of our proposed model is compared with the four baseline approaches, as shown in Table 5. In the baseline approach, some models, such as BERT-base and RoBERTa-base, use their powerful context representation capabilities to capture long-term dependencies in text and provide rich semantic information for sentiment analysis tasks. However, this single model is not effective in capturing local semantic patterns. Other models use a combination of Transformer and machine learning models. However, some models, such as Amobee, IIIDYT, and CEFER, have the problem of static feature fusion, and the attention weight is not dynamically adjusted, resulting in the dilution of key local features. There are also models like DataSEARCH and HUMIR, in which multiple modules extract features in parallel and independently, resulting in large performance fluctuations in some scenarios. All baseline methods are as follows:

- (1) Amobee (Rozental et al., 2018) uses a language model and an LSTM-based network and introduces a CNN attention mechanism for classification and prediction of emotions.
- (2) IIIDYT (Balazs et al., 2018) introduces ELMo and BiLSTM to improve the expressiveness of the model and uses the max-pooling technique to aggregate the LSTM output.
- (3) NTUA-SLP (Chronopoulou et al., 2018) is a framework for multi-task learning based on pre-trained models, integrating pre-trained models, multiple transfer learning strategies, and integrated learning methods for sentiment polarity classification tasks.
- (4) CEFER (Khoshnam et al., 2022) considers both sentence and individual word-level information and uses a combination of embedded sentiment feature vectors and contextually relevant feature vectors for sentiment analysis.
- (5) UBC-NLP (Alhuzali et al., 2018) uses BiLSTM, dropout regularization, and L2-norm regularization to optimize the model performance.
- (6) DuFER (Khoshnam et al., 2022) combines techniques such as sentiment modeling, classification, and hierarchical structures to improve the quality of the feature vectors by modifying the TF-IDF weighting and the maximum likelihood estimation weighting.
- (7) HUMIR (Naderalvojud et al., 2018) extracts context features, word features, sentiment weight features, and sentiment word embedding features, and then inputs the different types of features into a Multilayer Perceptron or multiple Convolutional Neural Networks for sentiment classification.
- (8) NLP (Zhou and Wu, 2018) uses a BiLSTM and an attentional mechanism for categorization.

Table 4
The results on the IEST dataset.

Methods	Parameters	ACC (%)	F1 (%)	Precision (%)	Recall (%)
Amobee	2024	71.48	71.45	71.53	71.37
IIDYT	3072	71.05	71.05	71.09	71.01
NTUA-SLP	2400	70.90	70.29	70.97	69.62
CEFER	4104	70.57	70.37	70.57	70.19
UBC-NLP	1724	69.66	69.28	69.70	68.87
DuFER	–	68.93	68.26	68.97	67.56
HUMIR	1224	68.80	68.64	68.85	68.43
NLP	1200	68.75	68.48	68.80	68.15
DataSEARCH	1000	68.66	68.04	68.71	67.38
EmotiKLUE	900	67.52	67.13	67.58	66.69
RoBERTa-base	768	67.03	67.01	67.05	66.97
BERT-base	768	65.02	64.91	65.07	64.75
RFGSA	2176	72.93	72.85	73.02	72.68

Table 5
The results on the EID dataset.

Methods	Parameters	ACC (%)	F1 (%)	Precision (%)	Recall (%)
CEFER	4104	95.16	95.08	95.21	94.95
DuFER	–	95.27	95.21	95.33	95.09
RoBERTa-base	768	90.04	90.04	90.07	90.01
BERT-base	768	88.36	88.13	88.42	87.84
RFGSA	2176	95.08	95.03	95.14	94.92

- (9) DataSEARCH (Senarath and Thayasivam, 2018) uses a combination of LSTM and CNN as the main feature extractor for implicit sentiment classification.
- (10) EmotiKLUE (Proisl et al., 2018) is a deep learning system that combines left–right context-independent representations of sentiment words with topic distributions from the LDA topic model.
- (11) RoBERTa-base (Liu et al., 2019) improves BERT in terms of optimizing the training process, increasing the training data, and tuning the hyperparameters, which effectively improves the performance in various natural language processing tasks.
- (12) BERT-base (Devlin et al., 2019) is a pre-trained language model that efficiently learns the semantic and contextual information of text to extract rich features for sentiment classification.

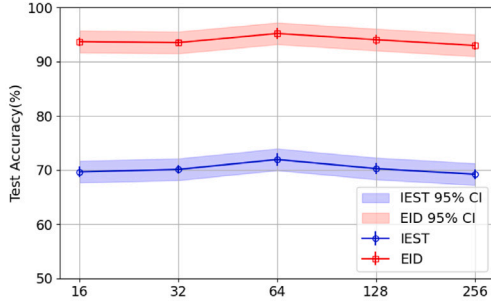
The experimental results on the IEST dataset are shown in Table 4. The results show that the accuracy of our model is improved by 1.45% to 72.93%, and the F1 score is improved by 1.4% to 72.85% compared to the Amobee model. The experimental results on the EID dataset are shown in Table 5. Although there is a slight decrease in performance on the EID dataset compared to the advanced model, the overall performance of the model remained within an acceptable range, demonstrating good generalization ability and stability. The reasons why our proposed model is superior to other models in the two datasets are as follows: (a) We designed a multi-level feature extraction model to enrich the semantic features of texts. The model uses the popular RoBERTa model for word embedding. Additionally, we apply the current advanced deep learning model to extract context and local features. This enables the model to understand and express text information more comprehensively. (b) We designed a multi-layer adaptive feature fusion scheme based on the gating mechanism. This scheme adaptively selects different levels of features for fusion and learns the weights of local and context features. It enables the model to adjust the importance and usage of features based on different types of texts.

In addition, it can be seen from the experimental results that the performance of all models on the IEST dataset was generally lower than on the EID dataset. This trend was consistent across all baseline model evaluation metrics. By nature, the IEST dataset contains less public sentiment information, relying on context and subtle cues to interpret emotions. This makes the model's task more challenging because it needs to rely on nuanced context understanding rather than explicit emotional markers. EID dataset have clear emotion words that provide a more direct signal to the model, making it easier to accurately learn and predict sentiment.

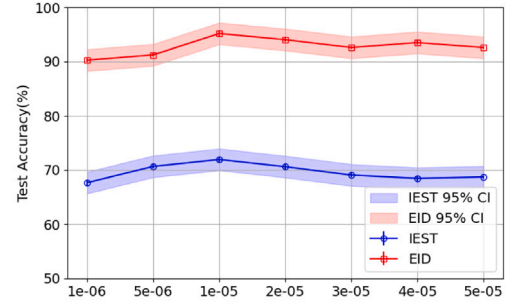
4.4. Hyperparameter sensitivity analysis

We analyzed the sensitivity of four hyperparameters through a series of hyperparameter experiments. These include Batch size, learning rate, output vector length, and dropout rate of feature fusion module, with Batch size set to values of 16, 32, 64, 128, 256, learning rate set to values of 0.000001, 0.000005, 0.00001, 0.00002, 0.00003, 0.00004, 0.00005, The output vector length is set to the values 1280, 1536, 1792, 2048, 2560 and the dropout rate of the feature fusion module is set to the values 0.01, 0.05, 0.1, 0.15, 0.2.

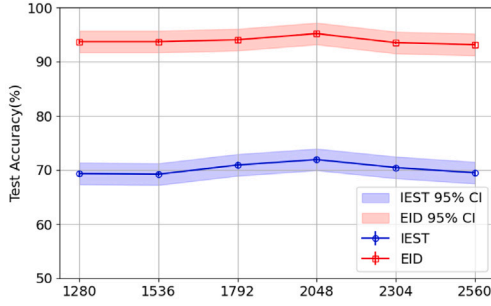
Fig. 4 shows the results of the hyperparameter sensitivity analysis. The results show that the model is less sensitive to batch size, dropout rate, and output vector length. The learning rate has a noticeable effect on the model's accuracy.



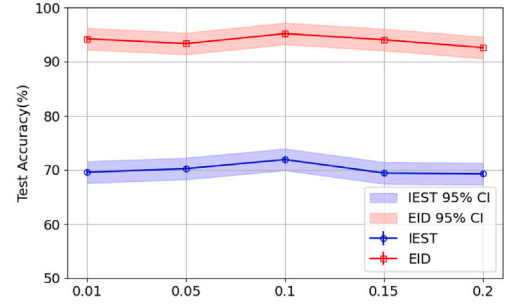
(a) Batch size



(b) Learning rate



(c) Output vector length



(d) Dropout rate of feature fusion module

Fig. 4. The results of the hyperparametric sensitivity analysis.

Table 6
Results of ablation experiments.

Models	IEST		EID	
	ACC (%)	F1 (%)	ACC (%)	F1 (%)
w/o RoBERTa	66.36	66.24	91.69	91.73
w/o Word	71.15	70.92	93.32	93.33
w/o Text	71.55	71.50	93.14	93.06
w/o GRU-FUSION	72.45	72.43	94.42	94.36
RFGSA	72.93	72.85	95.08	95.03

4.5. Ablation experiment

To validate the validity of each module in the model, we designed 4 variants of the RFGSA model. Ablation experimental results are shown in Table 6. “w/o RoBERTa” indicates using BERT instead of RoBERTa for word embedding representation. “w/o Word” refers to removing local Word-level features from the model. “w/o Text” model refers to removing the context characteristics from the model. “w/o GRU-FUSION” means that the model performs feature fusion directly by stitching, replacing the previous GRU-FUSION.

As shown in Table 6, the w/o RoBERTa model results are the worst of the three models, which indicates the importance of obtaining word embedding features based on RoBERTa. In addition, on the IEST dataset, the w/o Word model does not perform as well as the w/o Text model, however, it performs better on the EID dataset. It shows the validity of context features and local feature extraction in different scenarios. The performance of the w/o GRU-FUSION model is inferior to the RFGSA model, which indicates the necessity of an adaptive feature fusion module.

In addition, the feature distribution of different datasets is different. Theoretically, the effect of removing TextCNN or BiLSTM modules on model performance varies greatly on different data sets, while experimentally, there is little difference between the two. In general, local features usually refer to the vocabulary or phrase features that can be directly extracted from short texts, and these features play an intuitive and important role in sentiment analysis. For tasks that rely on explicit emotion words, local features are particularly critical. In the ISA dataset, because the emotion expression may be more dependent on the recessive features such as context and grammatical structure, the removal of local features may lead to greater performance fluctuation. Similarly, the BiLSTM model shows a similar trend. However, in some situations, a combination of tone, grammatical structure, or vocabulary may imply

Table 7
Results of performance experiments.

Models	IEST			EID		
	T	R	M	T	R	M
w/o RoBERTa	4.84	3.18	4731	4.81	3.54	4754
w/o Word	10.31	5.16	4770	10.26	6.09	4758
w/o Text	10.37	5.17	4763	10.35	6.18	4760
w/o GRU-FUSION	10.15	4.86	4736	10.07	5.45	4730
RFGSA	10.46	5.20	4788	10.39	6.23	4746

emotional information, and the importance of local features may be further enhanced. In this case, local features may be more important than contextual features.

4.6. Performance analysis

To evaluate the performance of the model, we analyzed the impact of each module on key metrics based on the ablation experiment in Section 4.4. We report three metrics:

T: Training speed per sample (ms/sample), computed by dividing the batch processing time by batch size 64.

R: Runtime speed per sample (ms/sample), calculated as the total processing time for 500 text samples divided by 500.

M: Peak memory usage (MB) during the inference task on 500 samples.

As shown in Table 7, the performance of the model during training and running is analyzed as follows:

(1) Training speed: In a single training iteration, the speed difference between the model variants (w/o Word, w/o Text, w/o GRU-FUSION, and RFGSA) was relatively small. The difference between the w/o GRU-FUSION model and the RFGSA model is the largest, about 0.3 ms/sample. This small difference suggests that the GRU-FUSION model has potential for further optimization in efficiency. Notably, the training speed of the w/o RoBERTa model is significantly faster than that of the other models. This is mainly due to removing the RoBERTa module, which simplifies the model structure and reduces the overall training cost. Under the same model configuration, the overall training speed shows minimal difference between the IEST and EID datasets, with the training speed for the EID dataset being slightly faster than that for the IEST dataset. This is influenced by two key factors: The EID dataset explicitly expresses sentiment, simplifying model processing and enhancing efficiency. In contrast, the IEST task involves a higher sentiment dimension. This increases the number of parameters, resulting in higher training costs.

(2) Runtime speed: During the model deployment inference phase, runtime performance on the IEST dataset is better than on the EID dataset. This indicates that the optimization of the model parameters in the training process may be more inclined to fit the characteristics of the IEST dataset. However, on the EID dataset, the accuracy and F1 score of the proposed model are similar to those of the DUFER model with a difference of less than 0.2%, which reflects the good generalization ability of the model.

(3) Memory consumption: The memory consumption of all model variants remained around 4.7 GB during run time (including training and inference), with negligible differences in real-time memory usage between models or across different datasets.

5. Conclusion

In this paper, we propose a RoBERTa-based sentiment analysis model RFGSA, and experimentally validate it on two public Twitter datasets. The experimental results showed that RFGSA showed significant advantages in handling both displayed text and implicit text sentiment analysis tasks in social media. On the WASSA 2018 IEST implicit sentiment dataset, our model improved accuracy and F1 values by 1.45% and 1.4%, respectively, over the best baseline model Amobee, and 4% and 4.59% over the DuFER model. In addition, on the Semeval 2018 explicit sentiment dataset, the accuracy and F1 score of the proposed model are similar to those of the DuFER model (difference < 0.2%), verifying the ability to generalize across tasks.

Compared with the traditional simple concatenation feature fusion method, the introduction of GRU mechanism in our proposed model inevitably increases the number of parameters and GRU requirements. However, in tasks that require high precision and fine-grained prediction, this increase significantly improves the model's performance, and the slight increase in computation time is effectively compensated for by the increase in accuracy. To further optimize this limitation, we plan to adopt a Sparse GRU based on L0 regularization, which is expected to significantly reduce the computational load by about 50%, thereby further improving the computational efficiency of the model.

Furthermore, this study focused on social media text features, excluding multimodal information such as images, audio, and video. To improve model performance in ISA and ESA, we plan to expand the data range to include multimodal data. Specifically, we plan to design a cross-modal sentiment analysis model that integrates text and image information. First, we transform the image modality into image captions. These captions are then fused with a global text encoder to extract global semantic features. To capture sentiment more accurately, we plan to design a fine-grained multimodal sentiment fusion module. This module extracts visual information related to aspects in images. Finally, we use MHCA to fuse visual features with text features. This approach combines both global and local semantic clues. It allows the model to better capture emotional information across text and images.

At the same time, we plan to conduct research on cross-language and cross-domain implicit sentiment analysis as a new direction for future studies. To improve cross-domain robustness, we will utilize adversarial domain-invariant training and pragmatic

consistency regularization. This will be evaluated on a newly curated benchmark, covering social media, product reviews, and conversational forums. For cross-language generalization, we plan to design a culturally adaptive language modeling approach. This approach will integrate language-specific adapter layers into multilingual transformers, such as XLM-R. We will then train these layers on linguist-annotated corpora, including sarcasm and indirect expressions in Chinese, Spanish, and Arabic.

CRedit authorship contribution statement

Xingyu Feng: Writing – original draft, Methodology, Investigation, Conceptualization. **Qindong Sun:** Validation, Supervision, Resources, Investigation. **Yan Wang:** Visualization, Validation, Methodology. **Yimin Qiao:** Visualization, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Abas, Ahmed R., Elhenawy, Ibrahim, Zidan, Mahinda, Othman, Mahmoud, et al., 2021. BERT-CNN: A deep learning model for detecting emotions from text. *Comput. Mater. Contin.* 71 (2), 2943–2961. <http://dx.doi.org/10.32604/cmc.2022.021671>, URL: <https://www.sciencedirect.com/science/article/pii/S1546221821001314>.
- Abdul, Rauf Saad, Yan, Qiang, Basit, Ali Syed, Waqas, Ahmad, et al., 2019. Using BERT for checking the polarity of movie reviews. *Int. J. Comput. Appl.* 975 (8887).
- Alhuzali, Hassan, Elaraby, Mohamed, Abdul-Mageed, Muhammad, 2018. UBC-NLP at IEST 2018: Learning implicit emotion with an ensemble of language models. In: *Proceedings of the 9th Workshop on Computational Approaches To Subjectivity, Sentiment and Social Media Analysis*. pp. 342–347.
- Ananthajothi, K., Karthikayani, K., Prabha, R., 2022. Explicit and implicit oriented aspect-based sentiment analysis with optimal feature selection and deep learning for demonetization in India. *Data Knowl. Eng.* 142, 102092.
- Anuratha, K., Parvathy, M., 2022. Twitter sentiment analysis using social-spider lex feature-based syntactic-senti rule recurrent neural network classification. *Internat. J. Uncertain. Fuzziness Knowledge-Based Systems* 30 (Supp01), 45–66.
- del Arco, Flor Miriam Plaza, Molina-González, M. Dolores, Ureña-López, L. Alfonso, et al., 2022. Integrating implicit and explicit linguistic phenomena via multi-task learning for offensive language detection. *Knowl.-Based Syst.* 258, 109965.
- Balazs, Jorge, Marrese-Taylor, Edison, Yutaka, Matsuo, 2018. Iiidy at iest 2018: Implicit emotion classification with deep contextualized word representations. In: *Proceedings of the 9th Workshop on Computational Approaches To Subjectivity, Sentiment and Social Media Analysis*. Association for Computational Linguistics(ACL), Brussels, Belgium, pp. 50–56.
- Basiri, Mohammad Ehsan, Nemati, Shahla, Abdar, Moloud, et al., 2021. ABCDM: An attention-based bidirectional CNN-RNN deep model for sentiment analysis. *Future Gener. Comput. Syst.* 115, 279–294.
- Bo, Pang, Lillian, Lee, Shivakumar, Vaithyanathan, et al., 2002. Thumbs up? Sentiment classification using machine learning techniques. In: *Proceedings of the 2002 Conference on Empirical Methods in Natural Language Processing. EMNLP 2002*, Association for Computational Linguistics, pp. 79–86. <http://dx.doi.org/10.3115/1118693.1118704>, URL: <https://aclanthology.org/W02-1011/>.
- Budhi, Gregorius Satia, Chiong, Raymond, Pranata, Ilung, et al., 2021. Using machine learning to predict the sentiment of online reviews: A new framework for comparative analysis. *Arch. Comput. Methods Eng.* 28, 2543–2566.
- Cai, Siqi, Yuan, Jingling, Li, Lin, 2023. A mutual implicit sentiment analysis model with bundle-aware contrastive learning. In: *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing. ICASSP*, pp. 1–5.
- Chau, Michael, Li, Tim M.H., Paul W. C. Wong, et al., 2020. Finding people with emotional distress in online social media: A design combining machine learning and rule-based classification. *MIS Q.* 44 (2), 933–955.
- Cheruku, Ramalingaswamy, Hussain, Khaja, Kavati, Ilaiah, et al., 2024. Sentiment classification with modified RoBERTa and recurrent neural networks. *Multimedia Tools Appl.* 83 (10), 29399–29417. <http://dx.doi.org/10.1007/s11042-023-16833-5>.
- Chronopoulou, Alexandra, Margatina, Aikaterini, Baziotis, Christos, et al., 2018. NTUA-SLP at IEST 2018: Ensemble of neural transfer methods for implicit emotion classification. In: *Proceedings of the 9th Workshop on Computational Approaches To Subjectivity, Sentiment and Social Media Analysis*. Association for Computational Linguistics(ACL), Brussels, Belgium, pp. 57–64.
- Devlin, Jacob, Chang, Ming-Wei, Lee, Kenton, et al., 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Association for Computational Linguistics(ACL), pp. 4171–4186.
- Diwali, Arwa, Dashtipour, Kia, Saeedi, Kawther, et al., 2022. Arabic sentiment analysis using dependency-based rules and deep neural networks. *Appl. Soft Comput.* 127, 109377.
- Fu, Yujie, Liao, Jian, Li, Yang, et al., 2021. Multiple perspective attention based on double BiLSTM for aspect and sentiment pair extract. *Neurocomputing* 438, 302–311.
- Gan, Chenquan, Feng, Qingdong, Zhang, Zufan, 2021. Scalable multi-channel dilated CNN-BiLSTM model with attention mechanism for Chinese textual sentiment analysis. *Future Gener. Comput. Syst.* 118, 297–309.
- Gao, Ziwen, Li, Zhiyi, Luo, Jiaying, et al., 2022. Short text aspect-based sentiment analysis based on CNN+ BiGRU. *Appl. Sci.* 12 (5), 2707.
- Hearst, Marti A., 1992. Automatic acquisition of hyponyms from large text corpora. In: *COLING 1992 Volume 2: The 14th International Conference on Computational Linguistics*.
- Hema, M.S., Maheshprabhu, R., K. Sudheer Reddy, et al., 2023. Prediction analysis for parkinson disease using multiple feature selection & classification methods. *Multimedia Tools Appl.* 82 (27), 42995–43012. <http://dx.doi.org/10.1007/s11042-023-15280-6>.
- Khoshnam, Fereshteh, Baraani-Dastjerdi, Ahmad, 2022. A dual framework for implicit and explicit emotion recognition: An ensemble of language models and computational linguistics. *Expert Syst. Appl.* 198, 116686.

- Khoshtam, Fereshteh, Baraani-Dastjerdi, Ahmad, Liaghatdar, M.J., 2022. CEFER: A four facets framework based on context and emotion embedded features for implicit and explicit emotion recognition. *arXiv preprint arXiv:2209.13999*.
- Klinger, Roman, Clercq, Orphée De, Mohammad, Saif, et al., 2018. IEST: WASSA-2018 implicit emotions shared task. In: *Proceedings of the 9th Workshop on Computational Approaches To Subjectivity, Sentiment and Social Media Analysis*. Association for Computational Linguistics(ACL), pp. 31–42.
- Li, Da, Rzepka, Rafal, Ptaszynski, M., et al., 2020. HEMOS: A novel deep learning-based fine-grained humor detecting method for sentiment analysis of social media. *Inf. Process. Manage.* 57 (6), 102290.
- Liang, Bin, Su, Hang, Gui, Lin, Cambria, Erik, Xu, Ruifeng, et al., 2022. Aspect-based sentiment analysis via affective knowledge enhanced graph convolutional networks. *Knowl.-Based Syst.* 235, 107643. <http://dx.doi.org/10.1016/j.knosys.2021.107643>, URL: <https://www.sciencedirect.com/science/article/pii/S0950705121009059>.
- LIAO, Weizhi, MA, Yaheng, C.A.O., Yiling, et al., 2021. Two-stage fine-grained text-level sentiment analysis based on syntactic rule matching and deep semantic. *IEICE Trans. Inf. Syst.* 104 (8), 1274–1280.
- Liu, Yinhan, Ott, Myle, Goyal, Naman, et al., 2019. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*.
- Lu, Qiang, Sun, Xia, Sutcliffe, Richard, et al., 2022. Sentiment interaction and multi-graph perception with graph convolutional networks for aspect-based sentiment analysis. *Knowl.-Based Syst.* 256, 109840.
- Luo, Wanbo, 2022. Research and implementation of text topic classification based on text CNN. In: *2022 3rd International Conference on Computer Vision, Image and Deep Learning & International Conference on Computer Engineering and Applications*. CVIDL & ICCEA, IEEE, pp. 1152–1155.
- Mickel, Hoang, Alija, Bihorac Oskar, Rouces Jacobo, et al., 2019. Aspect-based sentiment analysis using BERT. In: *Hartmann, Mareike, Plank, Barbara (Eds.), Proceedings of the 22nd Nordic Conference on Computational Linguistics*. Linköping University Electronic Press, Turku, Finland, pp. 187–196, URL: <https://aclanthology.org/W19-6120/>.
- Minqing, Hu, Bing, Liu, et al., 2004. Mining and summarizing customer reviews. In: *Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. KDD '04, Association for Computing Machinery, New York, NY, USA, pp. 168–177. <http://dx.doi.org/10.1145/1014052.1014073>.
- Mohammad, Saif M., Bravo-Marquez, Felipe, 2017. Emotion intensities in tweets. In: *Proceedings of the 6th Joint Conference on Lexical and Computational Semantics*. *SEM 2017, Association for Computational Linguistics(ACL), pp. 65–77.
- Mutlu, Begum, Sezer, Ebru A., 2023. Enhanced sentence representation for extractive text summarization: Investigating the syntactic and semantic features and their contribution to sentence scoring. *Expert Syst. Appl.* 227 (10), 120302.
- Naderalvojud, Behzad, Ucan, Alaettin, Sezer, Ebru Akcapinar, 2018. HUMIR at IEST-2018: Lexicon-sensitive and left-right context-sensitive BiLSTM for implicit emotion recognition. In: *Proceedings of the 9th Workshop on Computational Approaches To Subjectivity, Sentiment and Social Media Analysis*. pp. 182–188.
- Ocampo, Nicolás Benjamín, Sviridova, Ekaterina, Cabrio, Elena, et al., 2023. An in-depth analysis of implicit and subtle hate speech messages. In: *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*. Association for Computational Linguistics(ACL), pp. 1997–2013.
- Paramonov, Ilya, Anatoliy Poletaev, 2021. Adaptation of semantic rule-based sentiment analysis approach for Russian language. In: *2021 30th Conference of Open Innovations Association FRUCT*. IEEE, pp. 155–164.
- Proisl, Thomas, Heinrich, Philipp, Kabashi, Besim, et al., 2018. EmotiKLUE at IEST 2018: Topic-informed classification of implicit emotions. In: *Proceedings of the 9th Workshop on Computational Approaches To Subjectivity, Sentiment and Social Media Analysis*. pp. 235–242.
- Rahab, Hichem, Haouassi, Hichem, Laouid, Abdelkader, 2023. Rule-based Arabic sentiment analysis using binary equilibrium optimization algorithm. *Arab. J. Sci. Eng.* 48 (2), 2359–2374.
- Rozental, Alon, Fleischer, Daniel, Kelrich, Zohar, 2018. Amobee at IEST 2018: Transfer learning from language models. In: *Proceedings of the 9th Workshop on Computational Approaches To Subjectivity, Sentiment and Social Media Analysis*. Association for Computational Linguistics(ACL), pp. 43–49.
- Saraswathi, N., Rooba, T. Sasi, Chakaravarthi, S., 2023. Improving the accuracy of sentiment analysis using a linguistic rule-based feature selection method in tourism reviews. *Meas. Sens.* 29, 100888.
- Sarawagi, Sunita, 2008. Information extraction. *Found. Trends Databases* 1 (3), 261–377. <http://dx.doi.org/10.1561/1900000003>.
- Senarath, Yasas, Thayasivam, Uthayasanker, 2018. DataSEARCH at IEST 2018: Multiple word embedding based models for implicit emotion classification of tweets with deep learning. In: *Proceedings of the 9th Workshop on Computational Approaches To Subjectivity, Sentiment and Social Media Analysis*. pp. 211–216.
- Shuang, Kai, Yang, Qianqian, Jonathan, Loo, et al., 2020. Feature distillation network for aspect-based sentiment analysis. *Inf. Fusion* 61, 13–23.
- Songwen, Pei, Lulu, Wang, Tianma, Shen, Zhong, Ning, et al., 2019. DA-BERT: Enhancing part-of-speech tagging of aspect sentiment analysis using BERT. In: *Yew, Pen-Chung, Stenström, Per, Wu, Junjie, Gong, Xiaoli, Li, Tao (Eds.), Advanced Parallel Processing Technologies*. Springer International Publishing, Cham, pp. 86–95.
- Tan, Kian Long, Lee, Chin Poo, Anbananthen, Kalaiarasi Sonai Muthu, et al., 2022. RoBERTa-LSTM: A hybrid model for sentiment analysis with transformer and recurrent neural network. *IEEE Access* 10, 21517–21525.
- Vasileios, Hatzivassiloglou, McKeown Kathleen R., et al., 1997. Predicting the semantic orientation of adjectives. In: *35th Annual Meeting of the Association for Computational Linguistics and 8th Conference of the European Chapter of the Association for Computational Linguistics*. Association for Computational Linguistics, Madrid, Spain, pp. 174–181. <http://dx.doi.org/10.3115/976909.979640>, URL: <https://aclanthology.org/P97-1023/>.
- Wankhade, Mayur, Rao, Annavarapu Chandra Sekhara, Kulkarni, Chaitanya, 2022. A survey on sentiment analysis methods, applications, and challenges. *Artif. Intell. Rev.* 55 (7), 5731–5780.
- Wei, Jiyao, Liao, Jian, Zhenfei Yang, et al., 2020. BiLSTM with multi-polarity orthogonal attention for implicit sentiment analysis. *Neurocomputing* 383, 165–173.
- Wenxiong, Liao, Bi, Zeng, Xiuwen, Yin, Pengfei, Wei, et al., 2021. An improved aspect-category sentiment analysis model for text sentiment analysis based on RoBERTa. *Appl. Intell.* 51, 3522–3533.
- Wu, Peng, Li, Xiaotong, Ling, Chen, et al., 2021. Sentiment classification using attention mechanism and bidirectional long short-term memory network. *Appl. Soft Comput.* 112, 107792.
- Xiong, Xuanrui, He, Mengting, Li, Tianyu, et al., 2024. Adaptive feature fusion and improved attention mechanism based small object detection for UAV target tracking. *IEEE Internet Things J.* 11 (12), 21239–21249.
- Xiong, Haoqiang, Yan, Zehao, Wu, Chuhan, et al., 2023. BART-based contrastive and retrospective network for aspect-category-opinion-sentiment quadruple extraction. *Int. J. Mach. Learn. Cybern.* 14 (9), 3243–3255.
- Yang, Shanliang, Xing, Linlin, Li, Yongming, et al., 2021. Implicit sentiment analysis based on graph attention neural network. *Eng. Rep.* 53, 3109–3123.
- Yerramaneni, Sowjanya, Reddy, Sudheer K., 2025. A review on breast cancer detection using machine learning techniques. *Int. J. Data Min. Model. Manag.* 17 (2), 142–164. <http://dx.doi.org/10.1504/IJDMMM.2025.10065995>.
- Zhang, Tong, Gong, Xinrong, Chen, C. L. Philip, 2021. BMT-Net: Broad multitask transformer network for sentiment analysis. *IEEE Trans. Cybern.* 52 (7), 6232–6243.
- Zhang, Xiang, LeCun, Yann, et al., 2015. Text understanding from scratch. *arXiv:1502.01710*. URL: <http://arxiv.org/abs/1502.01710>. CoRR abs/1502.01710.
- Zhou, Qimin, Wu, Hao, 2018. NLP at IEST 2018: BiLSTM-attention and LSTM-attention via soft voting in emotion classification. In: *Proceedings of the 9th Workshop on Computational Approaches To Subjectivity, Sentiment and Social Media Analysis*. Association for Computational Linguistics(ACL), pp. 189–194.
- Zongxi, Li, Haoran, Xie, Gary, Cheng, et al., 2021. Word-level emotion distribution with two schemas for short text emotion classification. *Knowl.-Based Syst.* 227, 107163.